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ISMRM British & Irish Chapter | Iberian Chapter

Dynamic Inversion Times for 3D LGE Imaging, using a Blood-Focused Algorithm: Initial Patient Demonstration

Jack Allen [1,2], George Mathew [1], Miriam Conway [1], Sophie Jenkins [1], David Firmin [1,2], Jennifer Keegan [1], Sonya V. Babu-Narayan [1] and Peter Gatehouse [1].

[1] Royal Brompton and Harefield Hospital, London, UK.
[2] National Heart and Lung Institute, Imperial College London.

Introduction

- Inversion-recovery 3D Late-Gadolinium Enhanced (LGE) imaging maps myocardial scar in patients with Atrial Fibrillation (AF).
- Patients with AF may have 3D LGE scans to aid the planning of ablation therapy and to assess the resulting scar distribution.
- Navigator-gated 3D LGE with single R-wave gating reduces the total sequence duration compared to alternate R-wave gating.
- For single R-wave gating, arrhythmia in patients with AF causes ghosting and poor myocardial nulling due to incomplete longitudinal magnetisation (Mz) recovery between cardiac cycles.
- Dynamic Inversion Times (TIs)[1] improve image quality by adjusting TI according to the previous cardiac cycle duration.
- Reductions in blood ghosting were shown by phantom studies for a *blood-focused* dynamic TI algorithm[2].
- Here we performed an *in vivo* comparison of the original dynamic TI algorithm[1] with the patient-specific blood-focused dynamic TI algorithm[2], both using an efficient respiratory gating algorithm (CLAWS[3]).

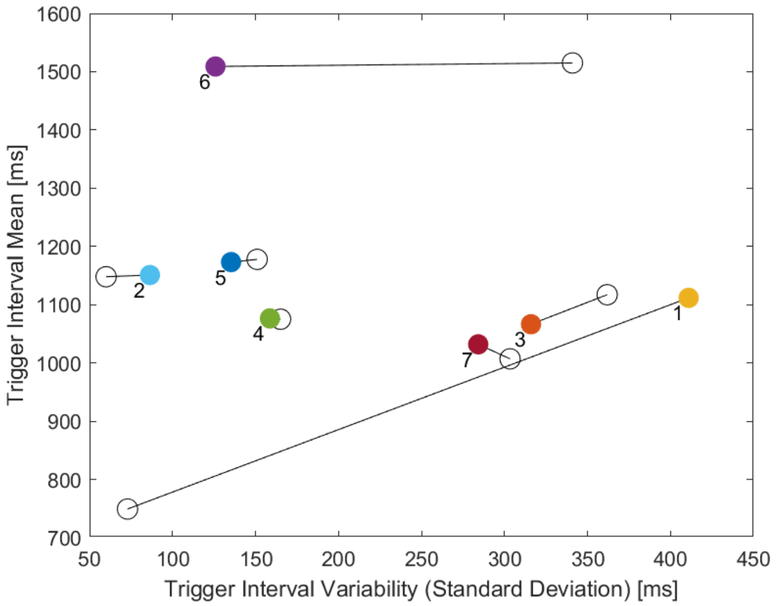

Figure 1: Trigger interval mean and variability for 7 patients (numbered), showing the *original* dynamic TI scan (clear circles) and the corresponding *blood-focused* scan (coloured circles).

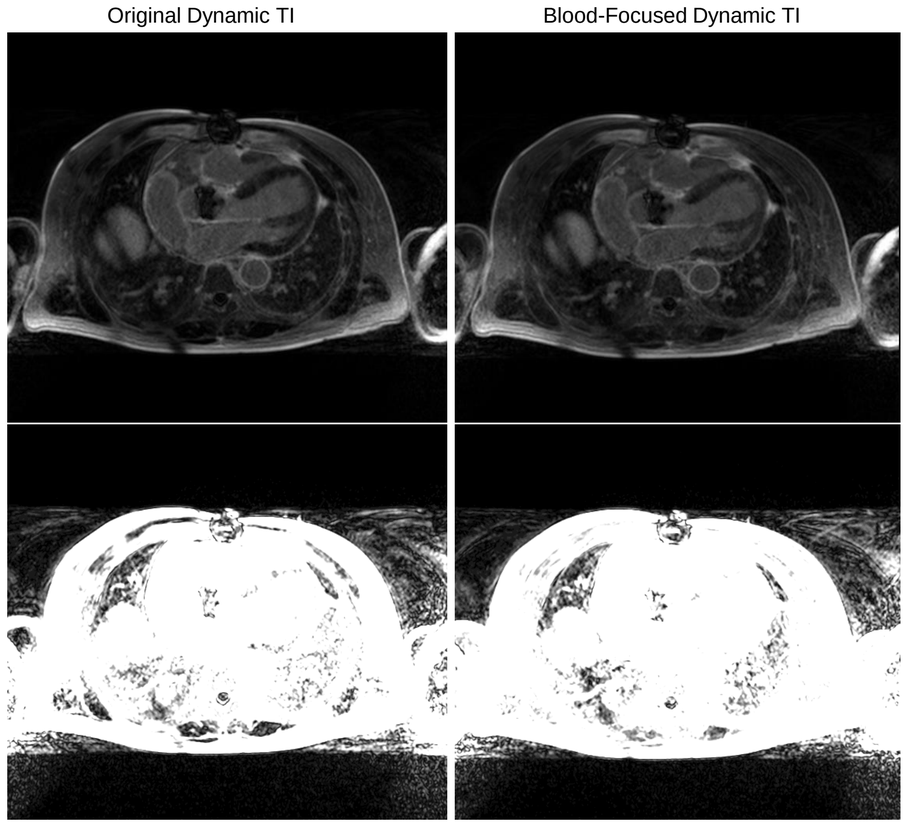

Figure 2: Central slice from patient 3, showing images from the two dynamic TI algorithms, with consistent greyscale ranges between the two. Top row: greyscale to show all tissue contrast. Bottom row: greyscale to highlight ghosting in the 'free space' outside the chest.

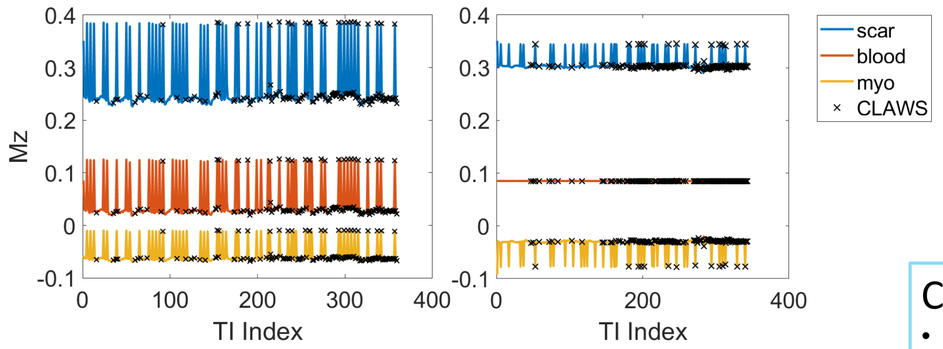

Figure 3: Bloch simulations of longitudinal magnetisation (Mz) for patient 3, for the original algorithm and blood-focused dynamic TI algorithm, showing consistent blood Mz across the TIs for the latter. Ideally, Mz for all tissue types would be consistent across all the TIs. The CLAWS algorithm uses the data from a subset of the TIs (crosses) to reconstruct the images.

Methods

Blood-focused algorithm:

- Uses the full history of the trigger intervals and sequence timing to model blood and myocardium longitudinal magnetisation (Mz) throughout the sequence.
- Uses blood T1, myocardium T1 and an estimation of the mean R-R interval to calculate TI in the current cycle to consistently achieve an unvarying target blood Mz, which is the blood Mz at the myocardium null time for the given mean interval.

Study:

- 3D LGE inversion-recovery scans were performed in 7 patients (Fig. 1) with congenital heart disease using two 3D LGE acquisitions: the original dynamic TI algorithm[1], followed by the blood-focused algorithm[2]. Example patient image in Fig. 2, with the simulated Mz in Fig. 3. Blood-ghost ratio was calculated using mean ghost and blood ROI measurements. A high blood-ghost ratio is desirable.

Results

- No significant change in mean ghosting (original: 7.1+/-2.4, blood-focused: 7.5+/-2.1, p>0.6).
- Small decrease in blood-ghost ratio for the blood-focused algorithm (original: 10.0+/-2.5, blood-focused 8.0+/-2.8, P<0.05, Fig. 4).

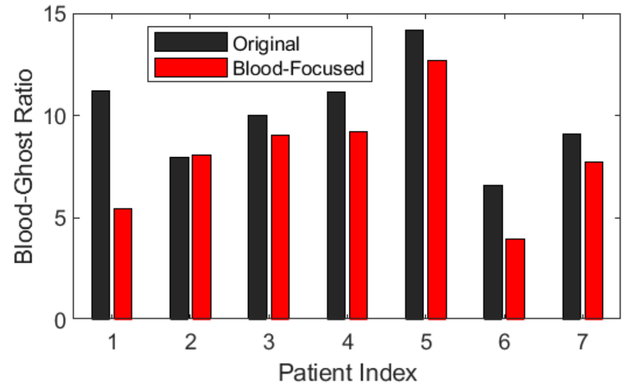

Figure 4: Blood-ghosting contrast across all patients.

Conclusions

- No improvement in *in vivo* image quality with the blood-focused dynamic TI algorithm[2]. Reduced blood-ghost contrast.
- Require more patients with large trigger interval variation.
- Important to swap the acquisition order in future scans to account for contrast-agent 'washout' which reduces blood signal intensity.
- Qualitative image scoring may be more meaningful for clinical used.

1. Keegan, J., et al. (2015), Magn. Reson. Med., 73: 646-654. **2**. Allen, J., et al. (2020), ISMRM. #2059. **3**. Jhooti, P., et al. (2010), Magn. Reson. Med., 64: 1015-1026.
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